Xezae Peshlakai

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SUMMARY

I am someone who strives to blend engineering with a sustainability-driven perspective, guided by my Navajo heritage and a commitment to building solutions that support both community and environment.
I am especially interested in solving real-world problems at the intersection of **power systems, hardware design, and clean energy**, where technical decisions directly shape efficiency, reliability, and long-term impact.
My strengths include hands-on system design, turning ideas into working solutions, and leading teams with a balance of technical depth, clear communication, and accountability.

EXPERIENCE

Intern Design/Electrical Engineering Intern

ArchKey Solutions — **June 2025 - September 2025, San Jose, California**

- Utilized **BIM** and **VDC methodologies** to support electrical design for large-scale commercial projects, improving **energy efficiency**, **constructability**, and **project lifecycle performance**.
- Applied Autodesk Revit and AutoCAD to develop and optimize electrical layouts, enhancing **system performance**, **load planning**, and **cost estimation accuracy**.
- Applied **energy-conscious engineering practices** to reduce **power consumption** and improve **system efficiency**, ensuring compliance with California Title 24 and adherence to National Electrical Code guidelines, including **proper conductor sizing**, **overcurrent protection**, **grounding/bonding**, and **safe installation practices** for high-performance building systems.

Supervisor

Rip Curl — **May 2022 - December 2024, Santa Cruz, California**

- Led daily retail operations by supervising team members, driving **operational efficiency**, and ensuring consistent execution of store objectives and customer service standards.
- Delivered high-quality customer experiences by applying **sales strategies**, resolving issues, and fostering strong **customer engagement** in a fast-paced retail environment.
- Managed **labor allocation**, **wages**, and **store expenses** in alignment with budget targets, while overseeing **inventory control** and **merchandising execution**.
- Supported store leadership in achieving **sales targets**, **revenue growth**, and **profitability goals** through team coordination and performance optimization.

Web Developer

Tech4Good — **September 2022 - March 2023**

- Developed responsive front-end components using Angular, translating Figma prototypes into production-ready interfaces with **HTML, CSS, and TypeScript**, improving **user experience** and interface consistency.
- Implemented back-end functionality using **Observables**, RxJS, and Firebase to build **scalable**, **data-driven web applications** with improved performance and reliability.
- Conducted research in **social computing**, analyzing the interaction between computational systems and human behavior to inform **user-centered design** and platform features.
- Contributed to the development of **Annota**, an AI-enhanced platform supporting **collaborative learning** and **qualitative analysis**, integrating **machine-assisted workflows** into web-based applications.

PROJECT

Solar Panel Cleaning Robot

Capstone Project • October 2025 - Present

- Designing an automated solar panel cleaning system to improve **solar asset performance**, **energy yield**, and **O&M efficiency** by reducing dust and debris accumulation.
- Integrating embedded control using a BeagleBone Black with motor drivers, sensors, and actuators to enable **autonomous operation**, **fault detection**, and real-time system control.
- Developing electrical schematics in KiCad, including **24V/12V/5V power distribution**, **power electronics interfaces**, **MOSFET switching circuits**, and **relay-based protection systems**.
- Performing component selection and system integration based on **voltage/current ratings**, **thermal constraints**, and **efficiency optimization**, ensuring reliable operation across mixed-voltage domains.
- Conducting electrical testing and validation using oscilloscopes and multimeters to evaluate **voltage regulation**, **load performance**, **transient response**, and implement **fault protection** and **safe shutdown mechanisms**.

EDUCATION

Bachelor of Science Electrical Engineering

University of California Santa Cruz • Santa Cruz, CA • 06/2026 • 3.54

- Member of College Scholars Program, Winter 2022 to present.

SKILLS

Energy & Electrical Systems: Power systems, **energy efficiency**, electrical system design, California Title 24 compliance

Embedded & Control Systems: Microcontrollers (STM32, PIC, BeagleBone Black), **real-time systems**, interrupts, sensor integration

Power Electronics & Hardware:DC-DC converters, motor drivers, battery systems, **MOSFET switching**, power distribution

Design & Engineering Tools: Autodesk Revit, AutoCAD, BIM/VDC, Onshape

Programming:C, embedded C, HTML/CSS

Leadership & Professional Skills:Team leadership, supervision, cross-functional collaboration, **project coordination**, problem-solving, communication, customer engagement, time management, adaptability